

Sai Shanmukh Habbar Kalle

 [linkedin.com/in/sai-shanmukh-habbar-kalle](https://www.linkedin.com/in/sai-shanmukh-habbar-kalle)
 github.com/shanmukh06

 saishanmukhkh@gmail.com
 [shanmukh06.github.io](https://github.com/shanmukh06)

EDUCATION

- University of Petroleum and Energy Studies** *2024 – Present*
 - B.Tech, Computer Science and Engineering (**8.24/10**) Dehradun, Uttarakhand
- Delhi Public School, GBN** *2023 – 2024*
 - 12th Grade, PCM (**79.9%**) Noida, Uttar Pradesh

RESEARCH INTERESTS

Speech Processing

EXPERIENCE

- National Institute of Technology Karnataka, Surathkal** *June 2026 – Present*
 - Research Intern Surathkal, Karnataka
 - Working under Prof. Deepu Vijayasenana on Speaker Profiling Tasks.

PROJECTS

ML/DL Paper Implementations — Independent Research *2024 – Present*

GitHub — Python, NumPy

- Algorithmic Reproduction:** Reproduced foundational ML/DL research papers from scratch using pure NumPy, including backpropagation and Adam optimization papers.
- Optimization Engine:** Implemented Adam and AdaMax optimizers with bias correction and numerical stability handling; validated on non-convex benchmarks such as the Beale Function.
- Neural Architecture Design:** Built modular multi-layer perceptron pipelines with manual forward/backward propagation, momentum-based gradient descent, and Xavier/He initialization.
- Benchmarking & Analysis:** Developed experiment suites for convergence visualization and optimization trajectory analysis; achieved **100% accuracy** on 6-bit symmetry detection benchmarks.

wtop — Cross-Platform System Monitor

2024 – Present

GitHub — Python, asyncio, psutil, rich

- Asynchronous Architecture:** Developed a high-performance CLI system monitor using **asyncio** and **psutil** for non-blocking real-time monitoring of CPU, RAM, and disk utilization.
- Performance Optimization:** Reduced monitoring overhead by **~30%** through efficient asynchronous polling and optimized sampling intervals.
- Resource Efficiency:** Achieved **<2% CPU footprint** during continuous monitoring on resource-constrained development systems.
- Modular Systems Design:** Architected a cross-platform codebase supporting Windows and Linux with decoupled data acquisition and CLI rendering.

SKILLS AND TOOLS

- Programming Languages:** Python, C
- Libraries & Tools:** NumPy, Pandas, Matplotlib, Git
- Systems:** Linux (Ubuntu)